



FABRIC STRUCTURE & DESIGN

Part: B

Course Code: FE 305-0723

Figuring with Extra Thread

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MAIN FEATURES

- Extra threads are added to *make decorative designs on fabric*.
- If the *extra threads are removed*, the basic ground fabric *still remains complete*.
- So, extra threads usually *do not reduce the basic strength of the fabric*.
- *Bright and contrasting colors can be used easily*.
- The amount of colored design can be changed without affecting the ground color.
- Produces clear and attractive designs.
- Allows strong color contrast.
- *Design area can be increased or reduced easily*.
- Better fabric strength than designs made only by long floats.

METHODS

Extra threads can be added as:

- Extra warp
- Extra weft
- Combination of extra warp and extra weft

Extra Warp Figuring

- Extra yarns are introduced in the warp direction.
- A separate warp beam is usually needed.
- This is because extra warp and ground warp may have different take-up rates.
- There are usually two or more sets of warp yarns with one set of weft yarns.

METHODS

Extra Weft Figuring

- Extra yarns are inserted in the weft direction.
- The loom must be able to insert more than one color or type of weft.
- Extra weft may be inserted continuously or only where the design is required.

Arrangement of Threads

Depending on the design, figuring and ground threads may be arranged as:

- 1 and 1
- 1 and 2
- 1 and 3
- Other suitable arrangements. The arrangement depends on the fabric structure and required design effect.

METHODS

Here, 1 and 1, 1 and 2, 1 and 3 describe how the figuring threads and ground threads are arranged in sequence.

- 1 and 1: 1 extra thread is followed by 1 ground thread.

Pattern: E – G – E – G – E – G

This gives a relatively strong and solid decorative effect.

- 1 and 2: 1 extra thread is followed by 2 ground threads.

Pattern: E – G – G – E – G – G

There are fewer figuring threads, so the design is less dense.

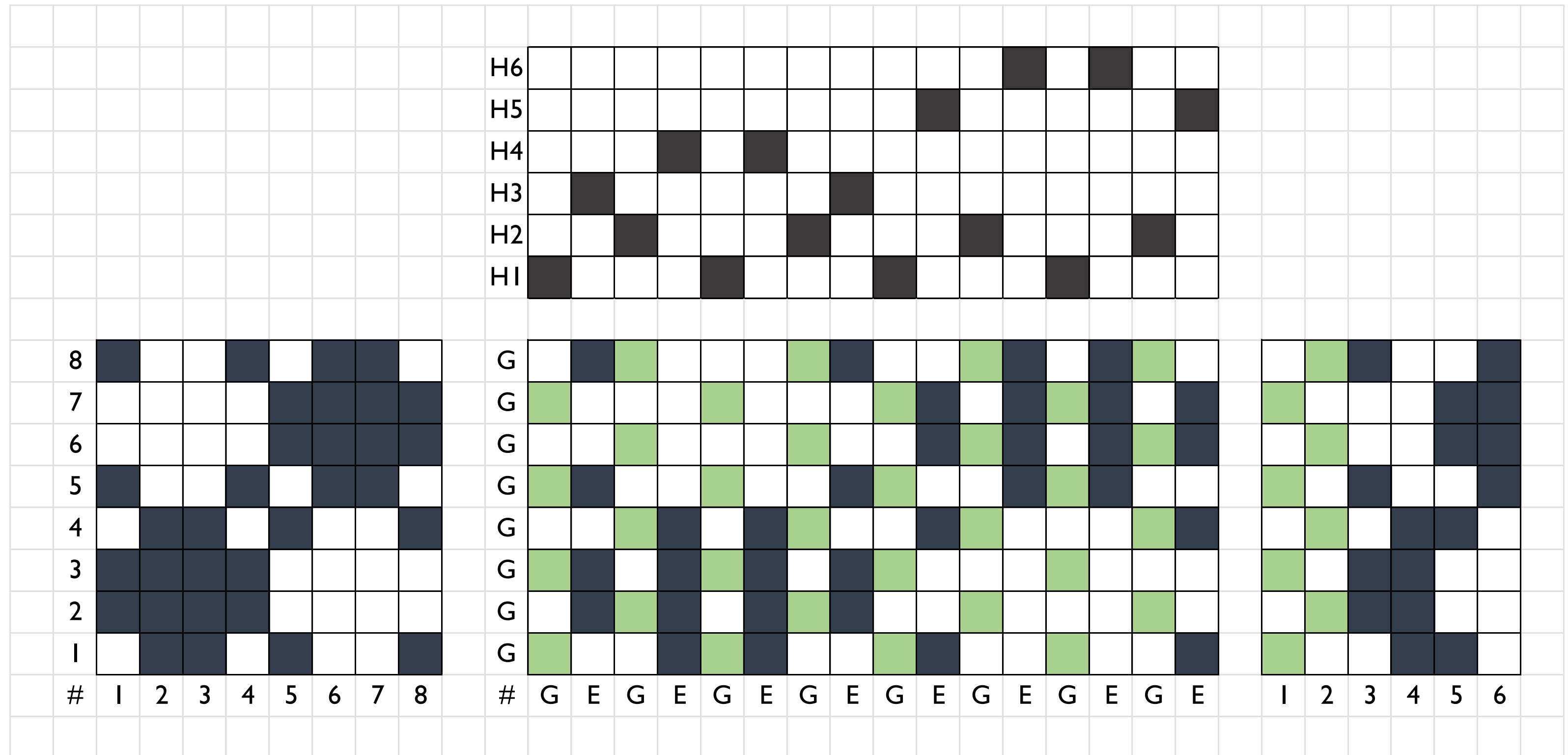
- 1 and 3: 1 extra thread is followed by 3 ground threads.

Pattern: E – G – G – G – E – G – G – G

The figuring yarns are more widely spaced, so the design is more open and uses less extra yarn.

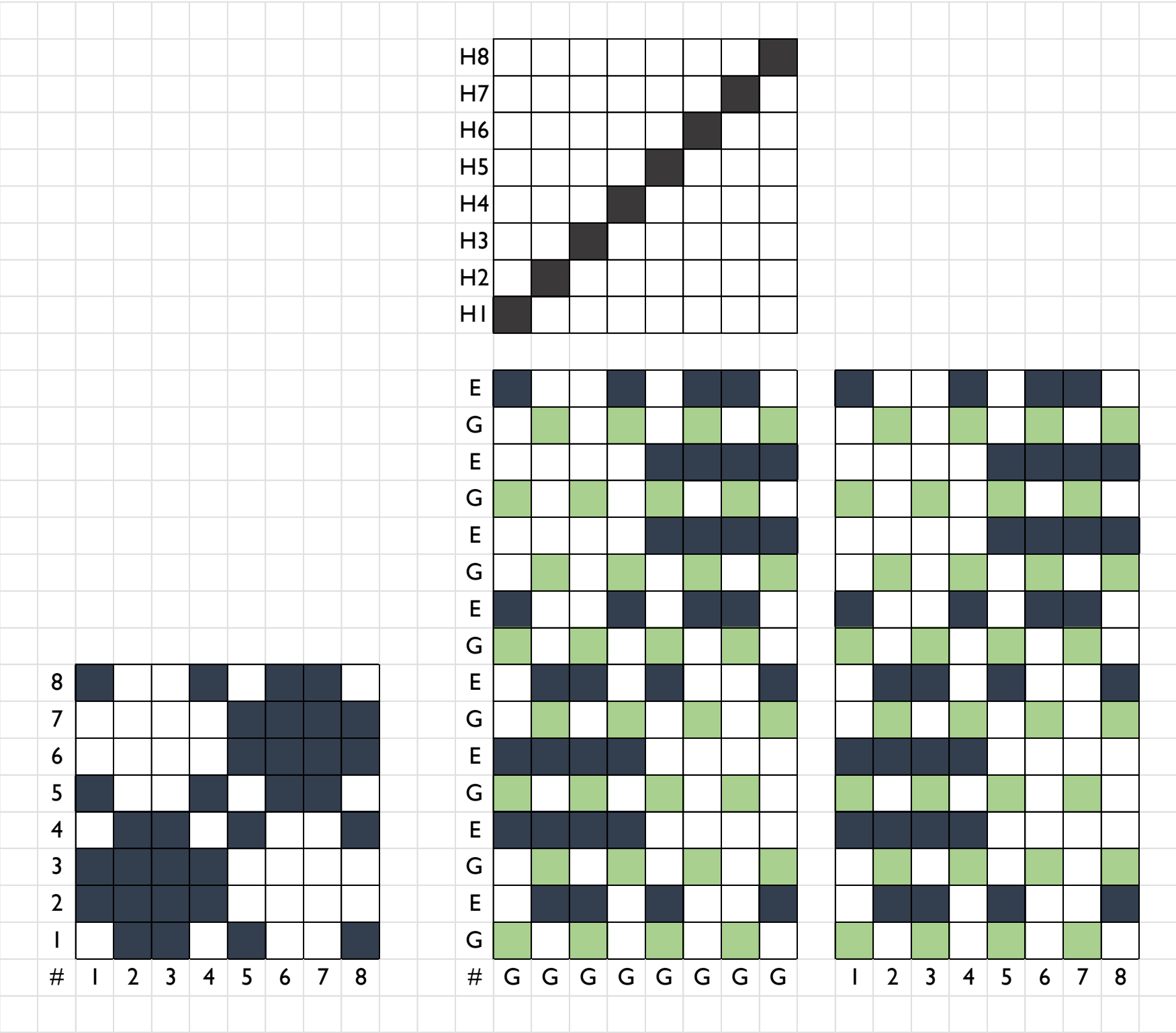
- Here, E = Extra thread and G = Ground thread.

EXTRA WARP DESIGN



EXTRA WARP DESIGN

EXTRA WEFT DESIGN



PROS OF FIGURING WITH EXTRA THREAD

1. *Higher loom productivity* because *only one set of weft picks is inserted*, so the loom can run faster.
2. *No special picking, shuttle box*, or take-up motion is needed.
3. *Many colors can be used*; theoretically, there is no fixed limit.
4. With an *intermittent arrangement of extra warp ends*, both *spot* and *stripe* designs can be made.
5. With a *similar arrangement* in extra weft, usually only *spot* designs are suitable, because *horizontal lines may look unattractive*.

CONS OF FIGURING WITH EXTRA THREAD

1. *Two or more warp beams may be needed instead of only one.*
2. With an ordinary Jacquard and harness, the design repeat width becomes smaller because extra warp ends need more harness space.
3. In dobby weaving, the draft becomes more complicated.
4. *Stronger yarn is needed for the extra warp* because it faces more tension during weaving.
5. *Extra warp yarns may be less soft, less full, and less shiny.*
6. Extra warp designs usually *look less prominent* than extra weft designs.
7. If extra yarns must be removed from the back of the fabric, *cutting extra warp ends is more difficult* and more *expensive* than cutting extra weft picks.

THANK
YOU !

